

## DETAILED DESCRIPTION OF THE INVENTION — DATA VISUALIZATION SYSTEMS AND NON-EXPLICIT REPRESENTATION FRAMEWORKS

Physiologic information generated by the intraluminal physiologic monitoring device may be communicated to users, clinicians, or computational systems through visualization architectures designed to convey insight while avoiding explicit anatomical representation.

In various embodiments, data visualization may employ abstract graphical formats including indices, scores, trend curves, symbolic representations, geometric patterns, color-coded indicators, or dynamically generated visual constructs derived from processed physiologic data. Visualization frameworks may prioritize interpretability, privacy preservation, and cross-cultural usability.

Abstract visualization systems may translate complex physiologic signals into simplified metrics representing states, trends, alignment conditions, variability signatures, or performance indicators. Representations may avoid depiction of anatomical imagery while maintaining meaningful interpretation capability.

Certain embodiments may generate symbolic visual identities derived from individualized physiologic profiles. Such representations may function as persistent visual summaries of longitudinal physiologic behavior and may be updated dynamically as new data becomes available.

Visualization environments may support interactive dashboards allowing users to explore historical trends, compare sessions, evaluate progress, or observe predictive forecasts generated by adaptive modeling systems. Interfaces may adapt to user preferences, expertise level, or clinical workflow requirements.

Multi-user visualization systems may enable comparison of abstracted profiles between authorized individuals without disclosure of raw physiologic measurements. Comparative displays may support compatibility evaluation, coaching applications, or collaborative analysis environments.

Visualization frameworks may extend across mobile applications, desktop platforms, wearable displays, augmented reality environments, virtual reality systems, or ambient display technologies integrated into everyday devices.

Privacy-preserving visualization techniques may intentionally obscure sensitive parameters while preserving relative relationships between metrics. Data abstraction layers may therefore act as protective intermediaries between raw physiologic data and user-facing interpretation.

The visualization and non-explicit representation systems described herein are illustrative rather than limiting. Any visualization approach capable of communicating physiologic insight while protecting user privacy is considered within the scope of the present disclosure.